

JINGYAO YU

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🌐 https://peteryu0131.github.io/Jingyao_Yu_Website/

EDUCATION

M.S. in Computer Science

University of Victoria
Sep 2025 – Present

B.S. in Computer Science

University of Victoria
Jan 2021 – July 2025

SKILLS

Programming Languages

C#, Python, Java, JavaScript, TypeScript, C, C++, SQL

Testing & Automation

Selenium, SpecFlow, Sikuli, Eggplant Functional, Regression Testing, Smoke Testing, E2E Testing, UI Testing, API Testing, Defect Tracking, Root Cause Analysis

Frameworks & Web/Mobile

.NET, WPF, Entity Framework, XAML, React, React Native, Expo, Node.js, WebSocket, HTML, CSS

Backend & Databases

Supabase, Appwrite, Oracle SQL, MySQL, SQLite, SQL Server

Machine Learning & AI

OpenAI API, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, MATLAB, Jupyter Notebook

Tools & Platforms

Git, GitHub, Bitbucket, Jira, Visual Studio, AWS, Docker, Railway, Twilio, Unity, EAS Build

CERTIFICATIONS

Google Cybersecurity Specialization

Jul 2025
ID: 6Z3RQ5XNJ6Y2

AWS Cloud Technology Consultant

Specialization
May 2025
ID: 43MGHGR37MYM

Machine Learning Specialization

May 2025
ID: FMNFR2UEL6DC

RECENT WORK EXPERIENCE

QA Analyst – Test Automation Engineer Co-op BC Liquor Distribution Branch — Burnaby, BC

May 2024 – Dec 2024

- Built automated regression, smoke, and end-to-end test suites using Selenium, Sikuli, SpecFlow, C#, and Oracle SQL in a .NET environment.
- Developed reusable automation libraries and helper methods to reduce duplicated test logic and improve framework maintainability.
- Designed the base automation architecture and testing strategy for the CARM project, supporting regression, smoke, E2E, and release validation.
- Created image-based UI automation scripts with Eggplant Functional to validate complex user workflows and improve test coverage.
- Investigated automation failures through logs, test reports, and root cause analysis to improve test reliability and release confidence.

Software Developer / QA Tester Co-op Eddyfi Technologies — Nanaimo, BC

Sep 2023 – Apr 2024

- Developed new desktop application features using C#, WPF, .NET, SQLite, Entity Framework, and City Framework.
- Designed and implemented UI workflows and report-related features, improving user experience and device data presentation.
- Built reporting functionality with Fast Report API and SSRS to generate structured device inspection reports.
- Developed a C# / WPF robot testing plugin to automate repeatable hardware feature testing and reduce manual validation effort.
- Tested and debugged desktop workflows, reproduced defects, analyzed logs/database issues, and verified fixes before release.
- Improved large-dataset performance using data virtualization and optimized IEnumerable traversal.

HIGHLIGHT PROJECTS

PolyFEM-Python – Differentiable FEM Simulation & ML Workflows

- Developed a Python API layer for PolyFEM to make C++ finite element simulation easier to use from Python.
- Designed structured simulation outputs for mesh data, displacement, stress, strain, and von Mises fields.
- Integrated differentiable FEM workflows with PyTorch and adjoint-based gradients for shape and material optimization.
- Built ML-ready workflows for batch simulation, parameter sweeps, dataset generation, and inverse design experiments.

MoodDump – AI Mood Tracking & Anonymous Community App

- Built a cross-platform mobile app with React Native, Expo, TypeScript, Supabase, and OpenAI API.
- Implemented mood check-ins, private journals, anonymous posts, image uploads, comments, tags, reminders, and mood analytics.
- Designed Supabase authentication, database schema, RLS access control, AI-generated daily replies, and EAS build release flow.

FlashCall – Automatic Call Assistant App

- Built an automated calling system with Node.js, WebSocket, Appwrite, Twilio, Google Speech-to-Text, JavaScript, and SQL.
- Implemented scheduled calls, DTMF menu navigation, real-time transcription, keyword detection, and automatic call transfer.
- Used Twilio callbacks to sync call status, database state, user permissions, and workflow recovery